



Wheel/Rail Damage Mechanisms

Richard Stock – Global Head of Rail Solutions

Plasser American / Plasser & Theurer

B O S T O N



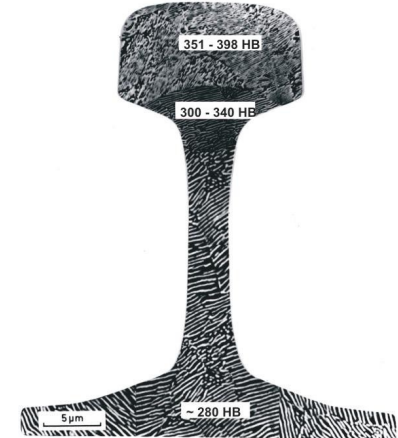
Materials

What Rails and Wheels Are Made Of

Steel: Rail and Wheel

- Rail: Hardness: 200 – 450 BHN
- Wheel: Hardness: 200 – 415 BHN
- Sufficient Toughness and Strength
- Ductile material behaviour
- Sufficient electric conductivity
- Reasonable weldability
- Excellent machineability
- Reasonable price

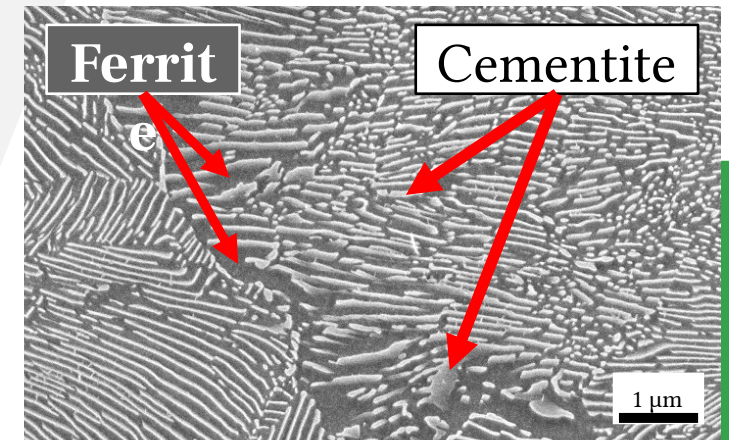
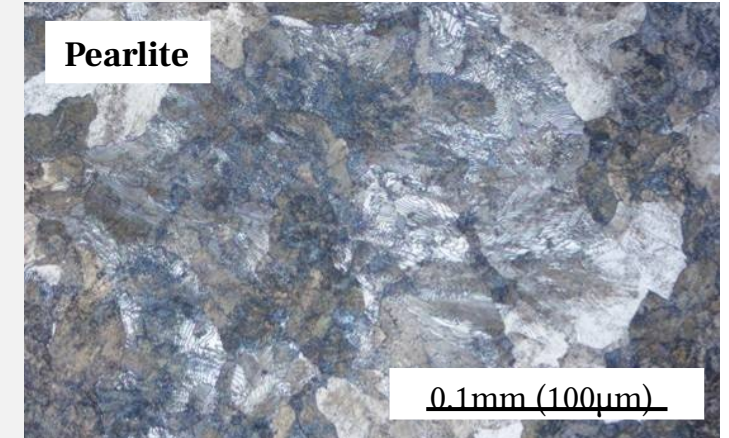
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Pearlitic Microstructure

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- Two phase material:
 - Ferrite: very soft, $C_{\max} = 0.02\%$
 - Cementite: Fe_3C , very hard, $C = 6.67\%$
- Lamellar or layer structure
- Pure pearlitic structure at 0.77% C (Eutectoid point)
 - $C > 0.77\%$: Hypereutectoid Steel
- Lamella spacing defines hardness and strength without influencing the toughness



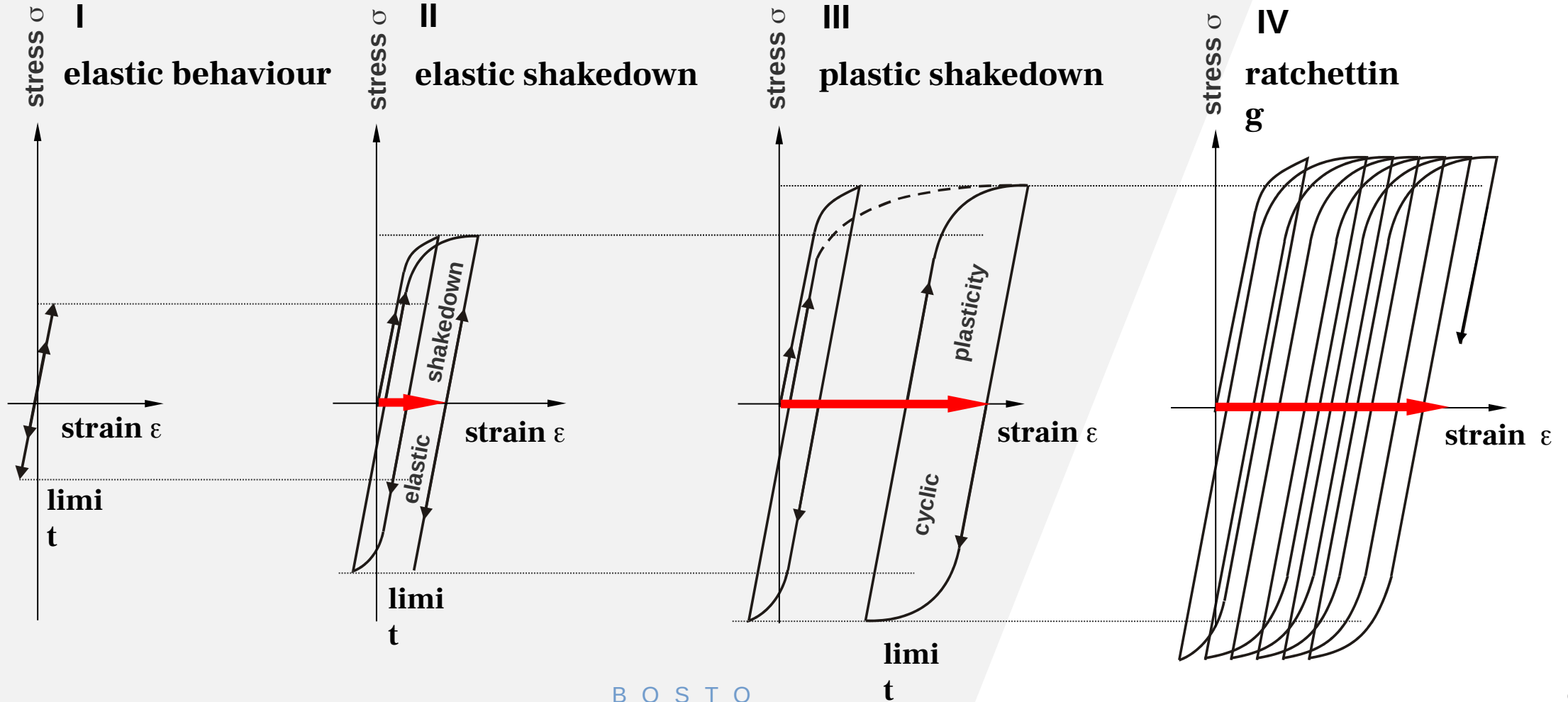


Wheel / Rail Damage Mechanisms

Material Reaction to Load

Material Behaviour Under Load

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Plastic Deformation

- Contact loads always above elastic material limit.
 - On a microscopic scale close to the rail/wheel surface
 - Incremental accumulation of plastic deformation - ratchetting
- Plastic flow enclosed by bulk elastic material
 - No “global” plastic flow

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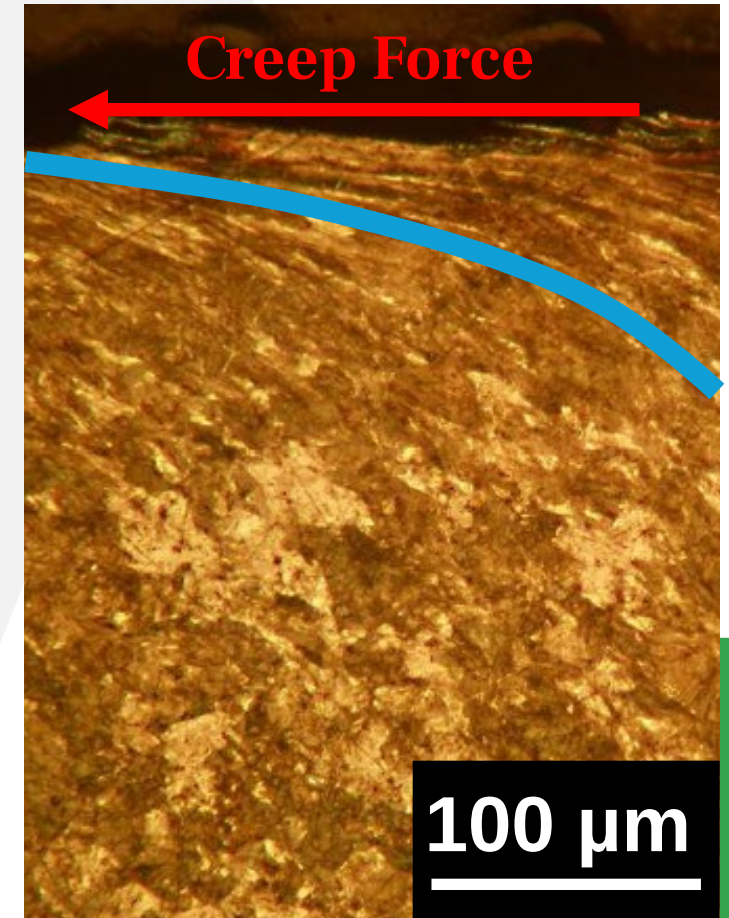
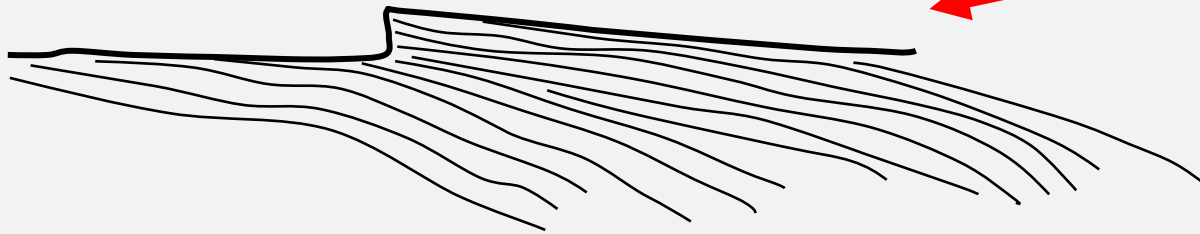


Photo: voestalpine

Material Response: Deformation

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Severely deformed and aligned material structure at the rail/wheel surface



Non-deformed material structure

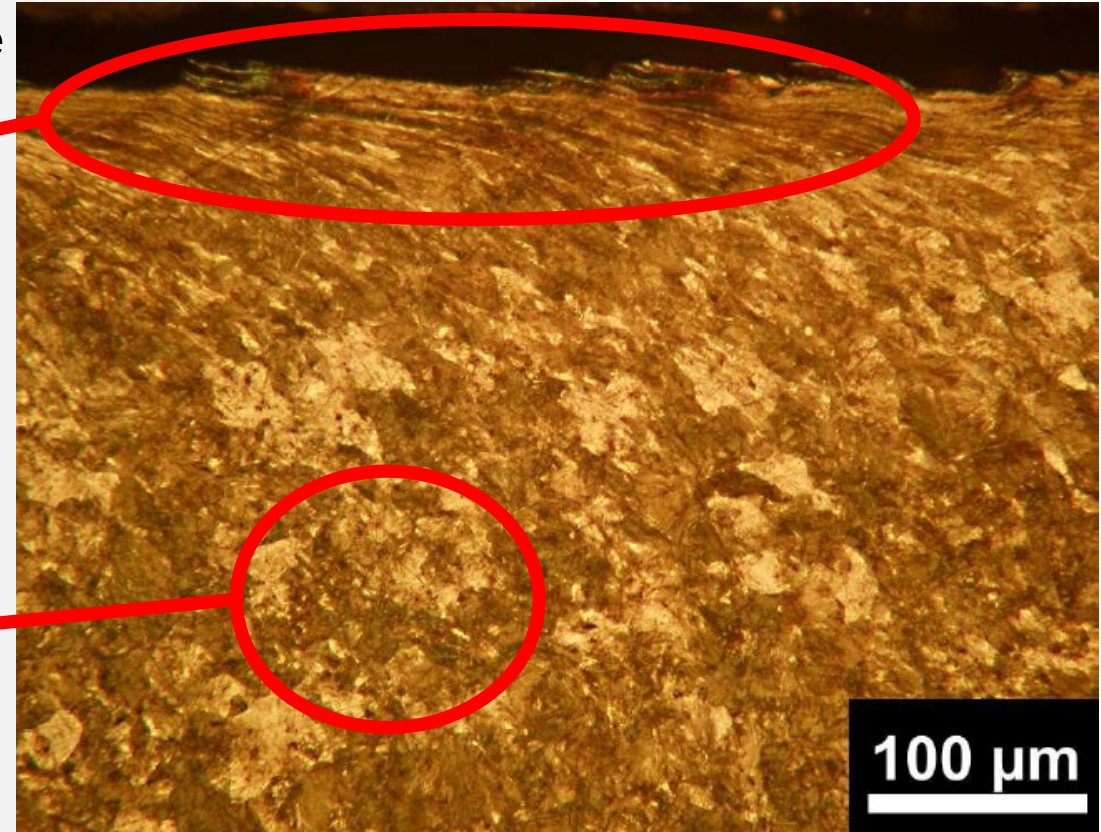
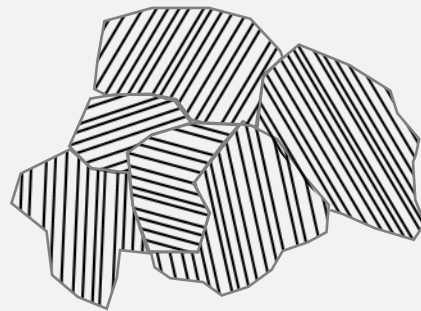


Photo: voestalpine

Plastic Deformation

- On a macroscopic scale – change of profile shape.
- Material flow – e.g. lipping

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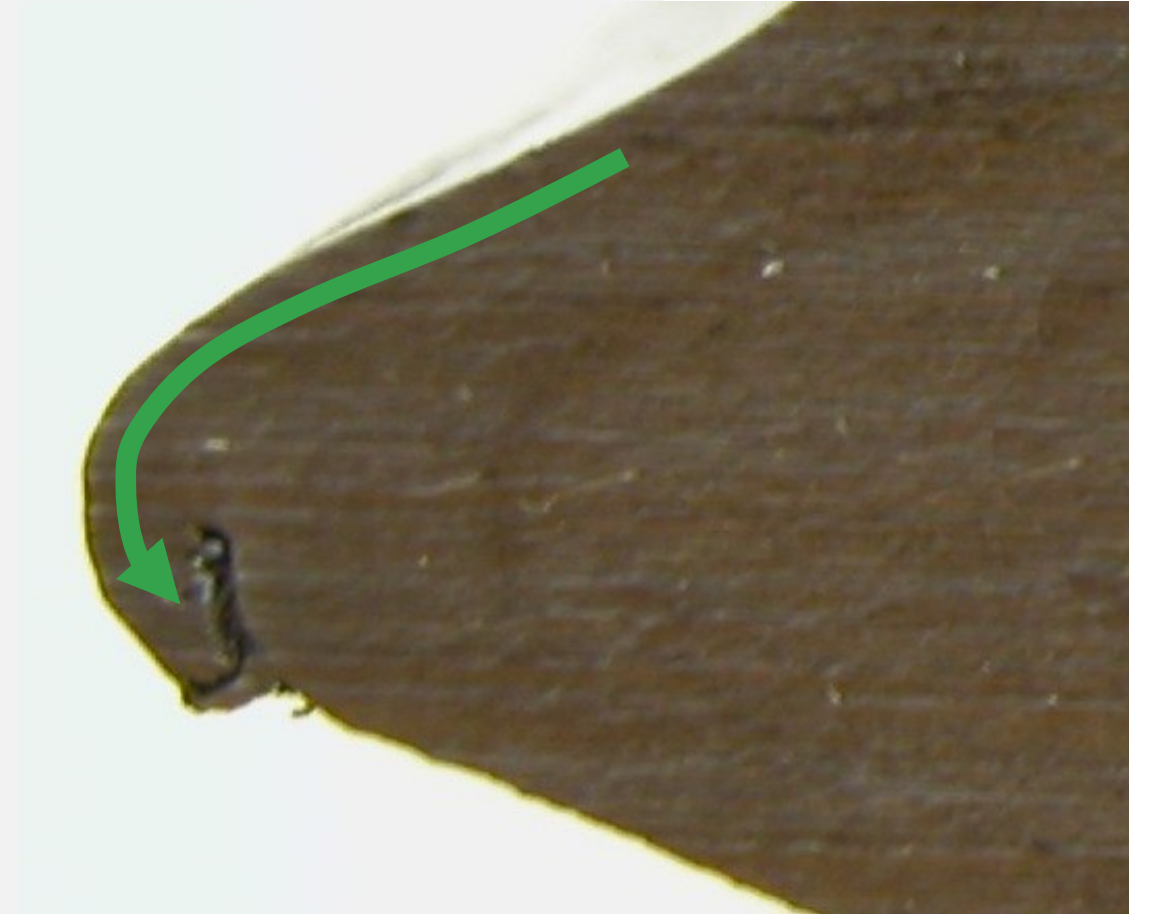


Photo: voestalpine

Material Wear

- Continuous material removal from rail/wheel surface due to interaction of wheel and rail.
- Several modes of wear
 - Adhesive wear
 - Abrasive wear
 - Fatigue wear
 - Corrosive wear
- Several types of wear
 - Natural Wear + Artificial Wear = Combined Wear

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Photo by L.B. Foster

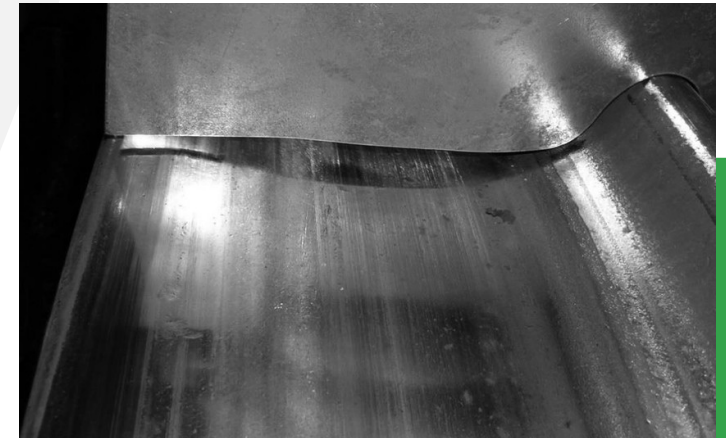


Photo by A. Eckberg

Material Response: Wear

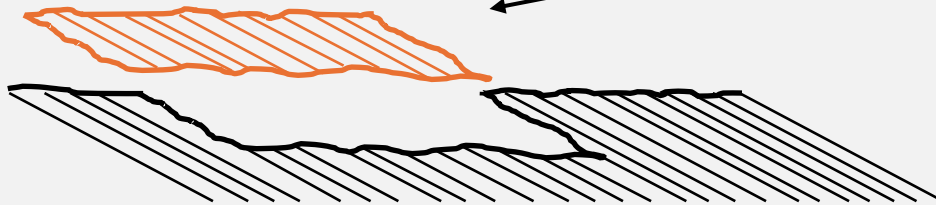
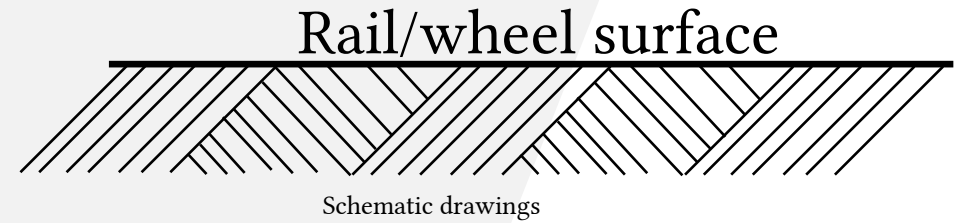
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Non-deformed, initial material condition

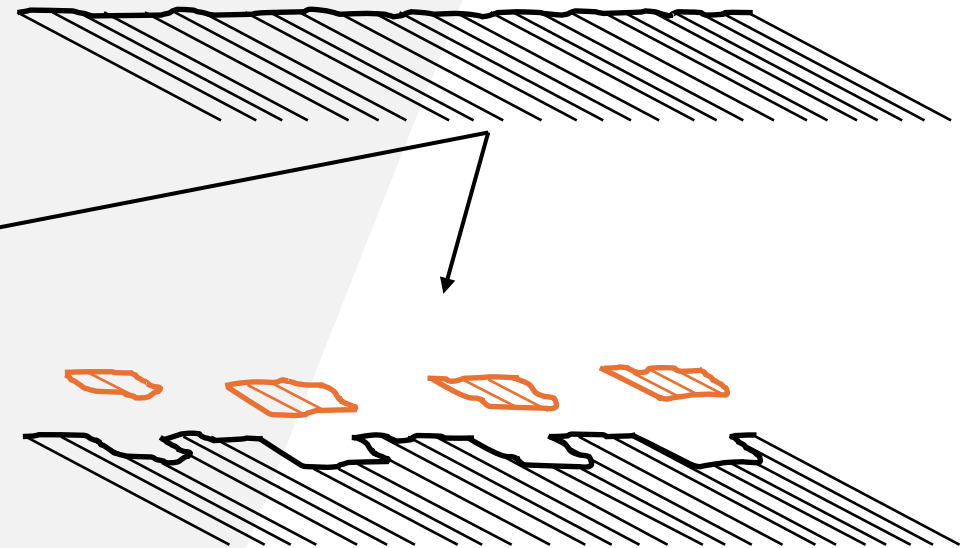


Loading conditions, material properties

Severely deformed rail/wheel surface



Severe wear

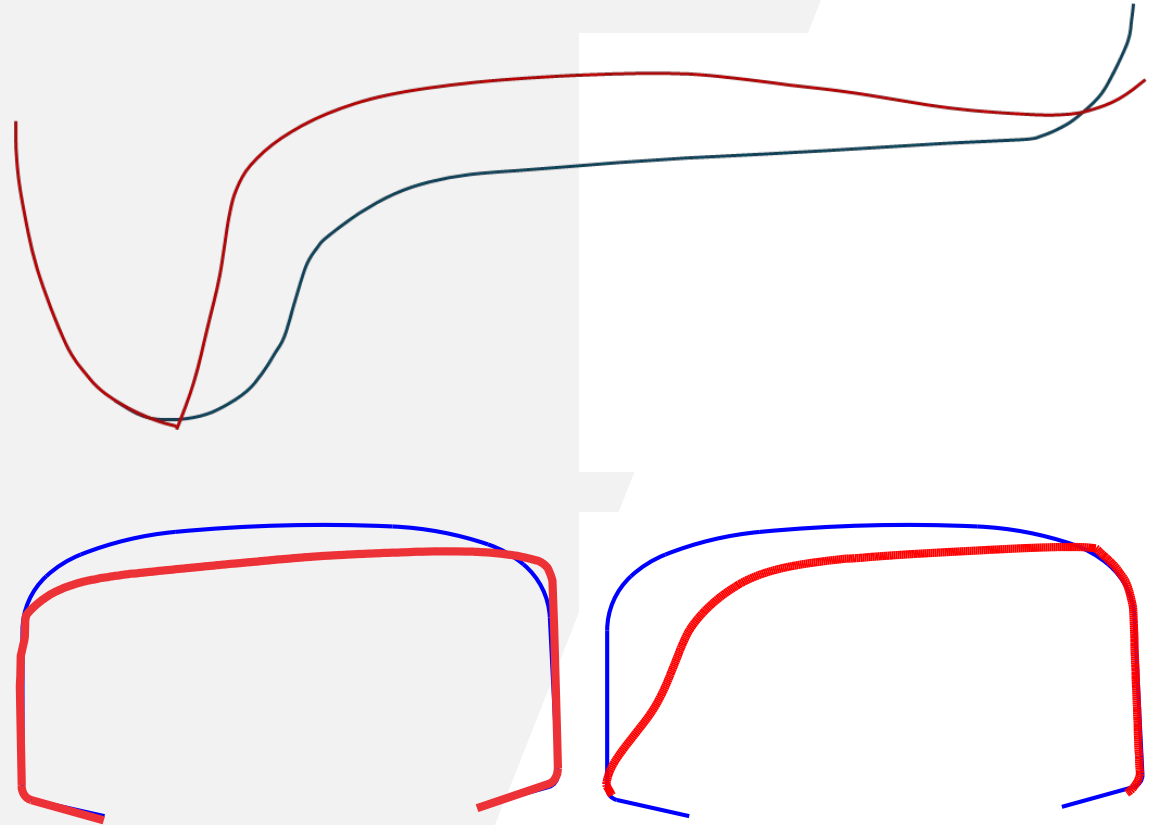


Mild wear

Wear effects: Profile Degradation

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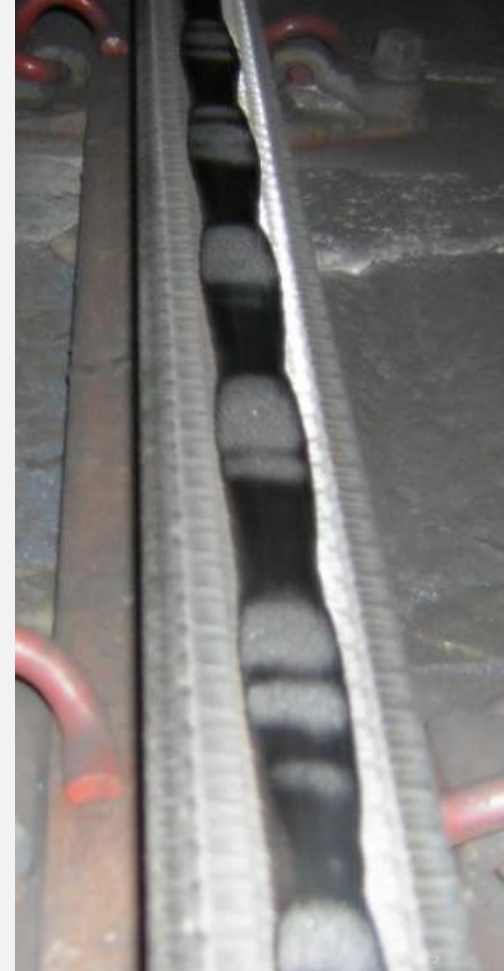
- Wheel:
 - Hollow Worn Wheel
 - Thin Flange
- Rail:
 - Flat Rail
 - Sever Gauge Wear



Rail: Corrugation

- Combination of wear and plastic flow
- Wave structure on the rail surface (tangent / curve)
- Short wave (25mm-80mm wavelength) or long wave (100-300mm) corrugation
- Multiple sub-classifications

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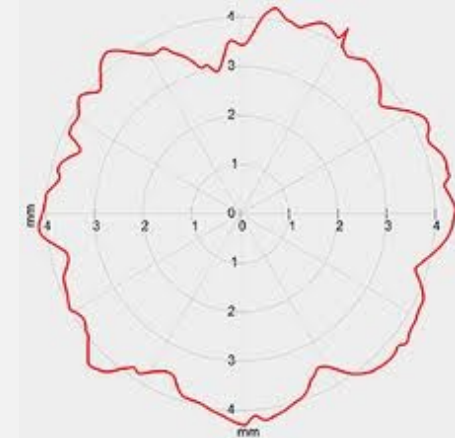


Photos by L.B. Foster

Out of Round Wheels

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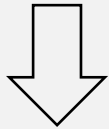
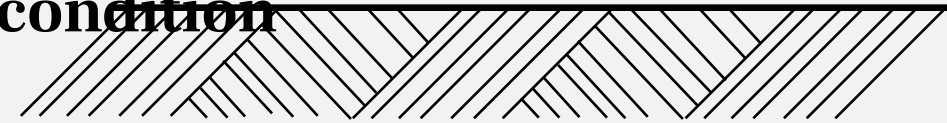
- Any deviation from a perfectly round wheel
- Polygonised wheels (periodic) caused by:
 - Stick-Slip Effects
 - Re-Profiling
 - Dynamic wheel-set oscillations caused by resonances in vehicle/track interaction.
- Wheel flat (discrete)
 - Wear during sliding of a locked wheel



Material Response: Cracks

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Non-deformed, initial material condition



Severely deformed surface

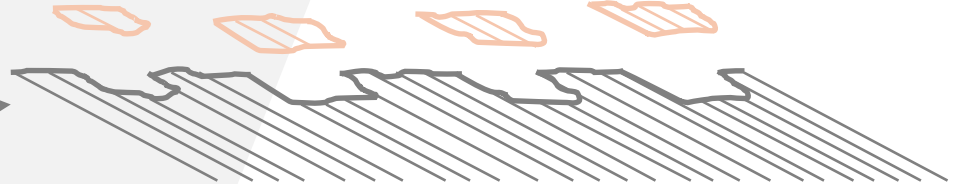


Schematic drawings

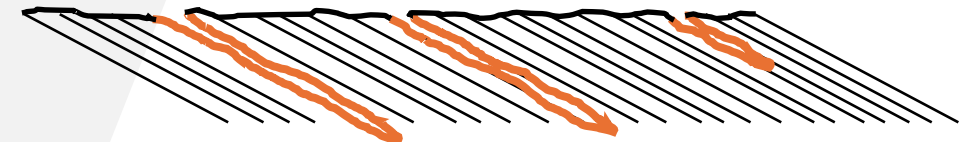
Severe wear



Mild wear

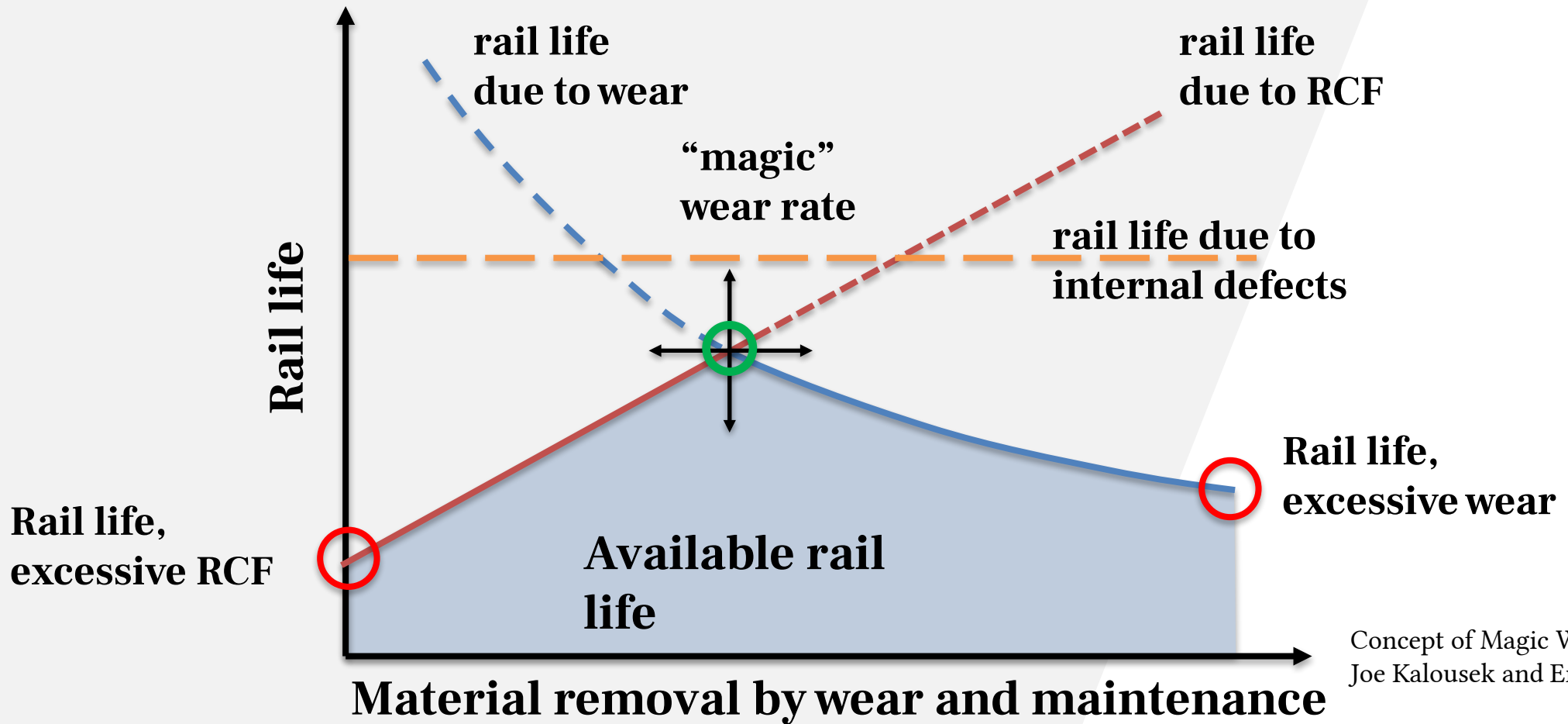


Surface cracks



Magic Wear Rate (Rail)

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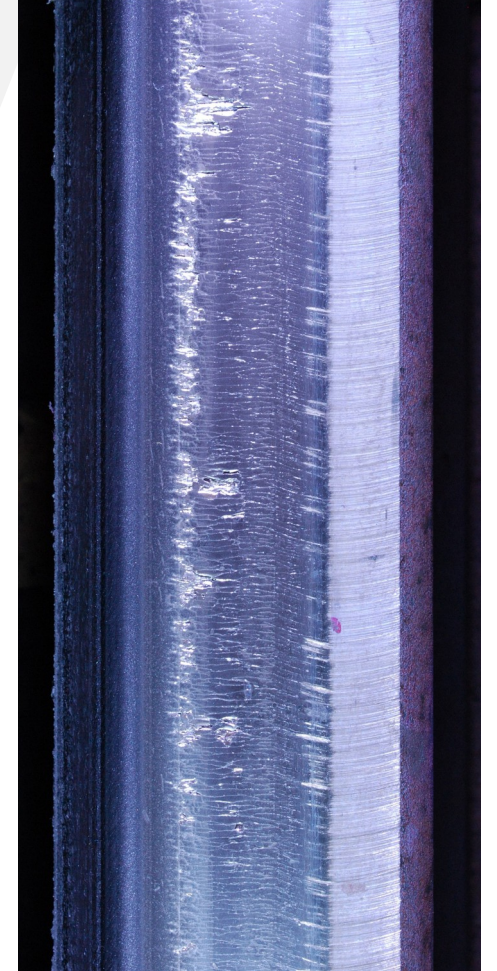


Concept of Magic Wear Rate by
Joe Kalousek and Eric Magel, 1997

Rail: Periodic Cracks

- Head Checks: periodic cracks at the gauge corner (gauge corner cracking)
- Sometimes: periodic cracks and crack networks also on the running surface
- Can cause detail fracture if not treated

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Wheel: Periodic Cracks

- Rolling Contact Fatigue Cracks
- At various positions along the wheel profile
 - Flange side to field side

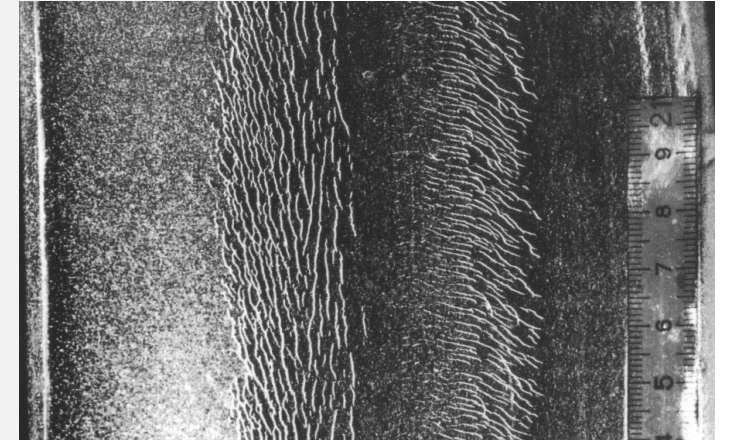
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Source: E. Magel and J. Kalousek, 2004



Source: Dedmon, CM2009



Source: E. Magel and J. Kalousek, 2004



Source: E. Magel and J. Kalousek, 2004

Rail: Flaking and Spalling

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- Head Checks can combine causing material to break out of the rail surface.
- Head Checks – Flaking – Spalling



Rail: Shelling

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- Originates underneath the rail surface
- Delamination of rail material – crack will surface at gauge corner and cause break-outs
- High loading conditions favor formation – overstressing of subsurface area



Wheel: Shelling

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- Different definition than rail
- Propagating cracks link beneath surface causing a large piece of material to spall off
- RCF cracks
- Thermo-mechanical (RCF) cracks
 - Material softening due to repeated heating



Source: E. Magel and J. Kalousek, 2004



Source: Eckberg and Kabo, 2005

Rail: Squats

- Widening of running band / dip
- Typical kidney shaped
- Surface and subsurface crack(s)
- Singular or massed occurrence
- Characteristics
 - Heavily sheared rail surface
 - Crack initiation and growth by ratcheting (RCF)
 - Slow growth (within 100 MGT)
 - Can result in rail break

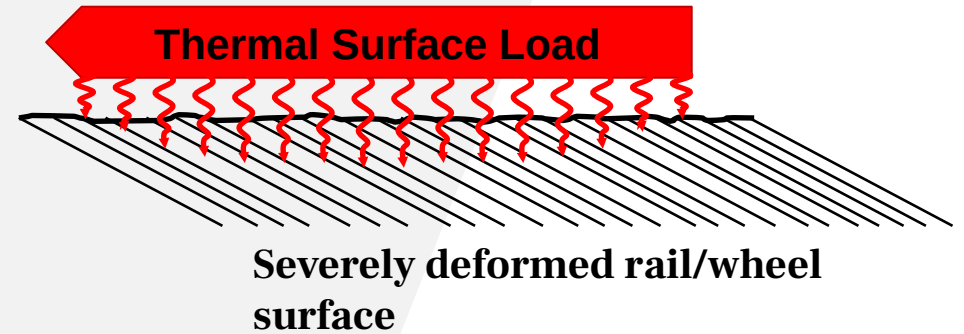
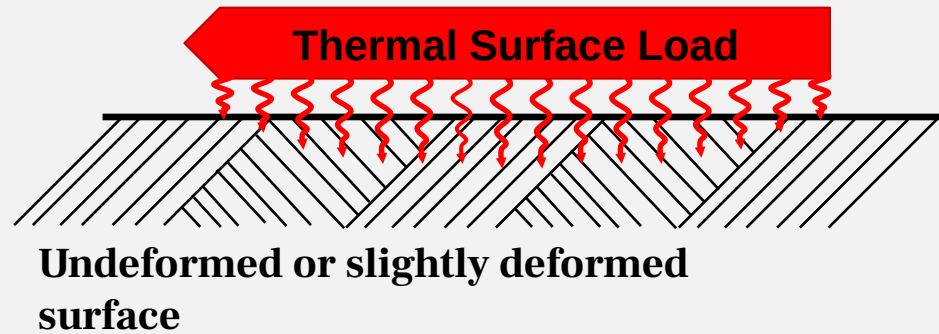
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Photos by voestalpine

Material Response: Thermal Transformation

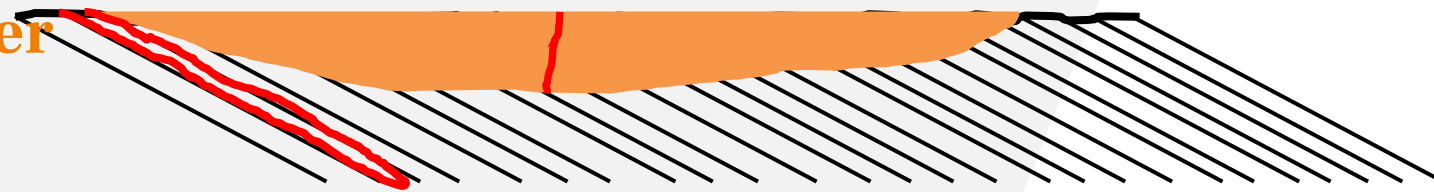
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Quenching by rail/wheel mass (heatsink)



Material Transformation: White/Brown Etching Layer



Cracks might develop at interface and/or within layer

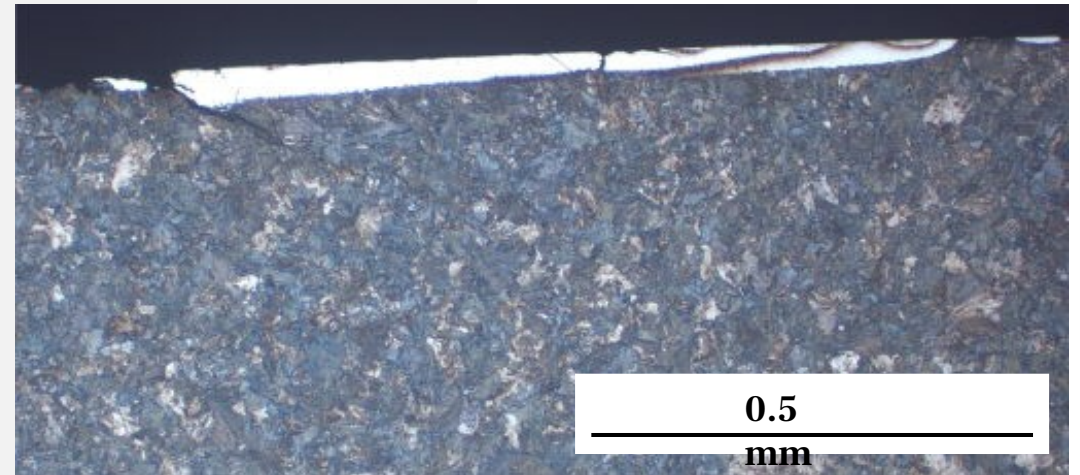
Schematic drawings

B O S T O

Thermal Transformation

- Thermal transformation of rail or wheel surface
- Source: traction systems, rail maintenance, braking
- Usually wears away
- White color in light microscope (LIMI)
 - Etching of surface to create contrast
 - 3% Nital Acid
 - Does not etch the Martensite / hard layers – appear white in LIMI

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Rail: Studs/ Squat Type Defects

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- Superficial similarity to Squats
- Multiple names
 - Studs, Squat Type Defects, Serial Squats, Squats
- Mostly epidemic appearance
- Extended spalling of rail surface possible



Photo: Rene Heyder, DB

Factors for Stud Development

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- Characteristics:
 - Almost no plastic deformation (no ratchetting)
 - Associated with “white etching layers” (martensitic layers)
 - Formation within 10MGT or less
 - Low wear conditions
 - Stress concentration by profile shape, grinding roughness or grinding facets
 - More prevalent with AC traction



Studs – A Complex Problem

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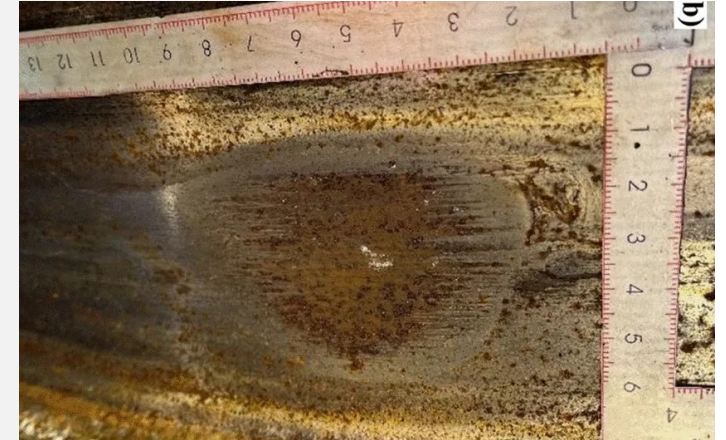
- Only/mostly outdoor sections affected
- No direct rail breaks reported
 - Exception: Interaction with HC/GCC
- Ongoing research and scientific debate
 - Disagreement on differentiation and naming



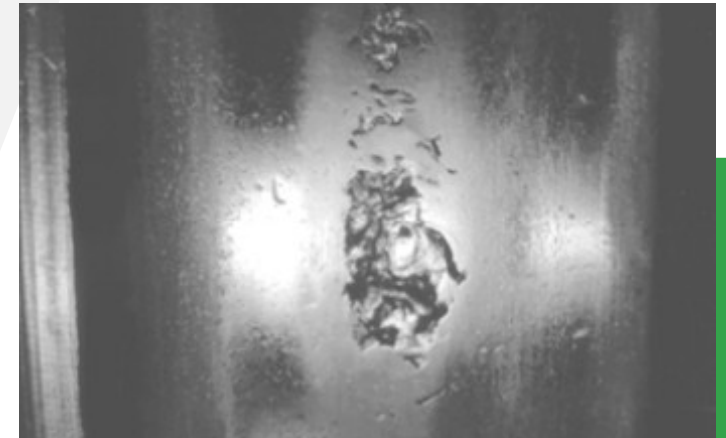
Wheel: Spalling

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- Different definition than rail
- Martensite patch induced by wheel sliding (skid flat) – thermal damage
- Crack initiation and material to break out of surface – Spalling
- Boundary between Shelling (RCF) and Spalling (thermal damage) blurry



Source: Maglio et al, 2024



Source: Magel and Kalousek, 1996

Rail: Wheel Burn

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- Occurs in pairs (both rails)
- Continuous slipping of locomotive wheel set(s).
- High temperature input to rail surface.
- Wear, plastic deformation, material transformation, cracks, break outs





Wheel: Wheel Flat

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- Wheel sliding along the rail
 - Counterpart to Wheel Burn
- Wear, plastic deformation, heat induces material transformation (martensite), cracks, spalling



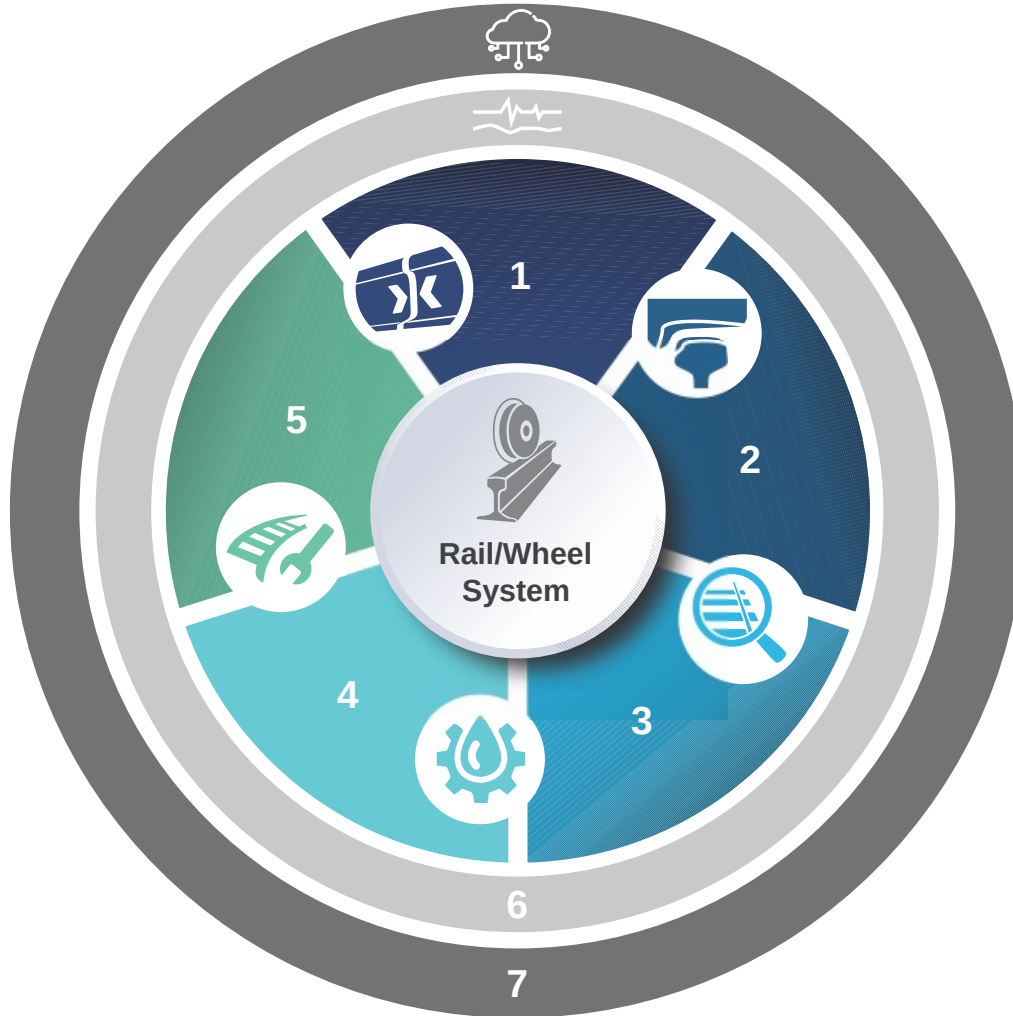
Source: Wikipedia, Bobo11, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons



Controlling Rail and Wheel Damage

How to manage asset life

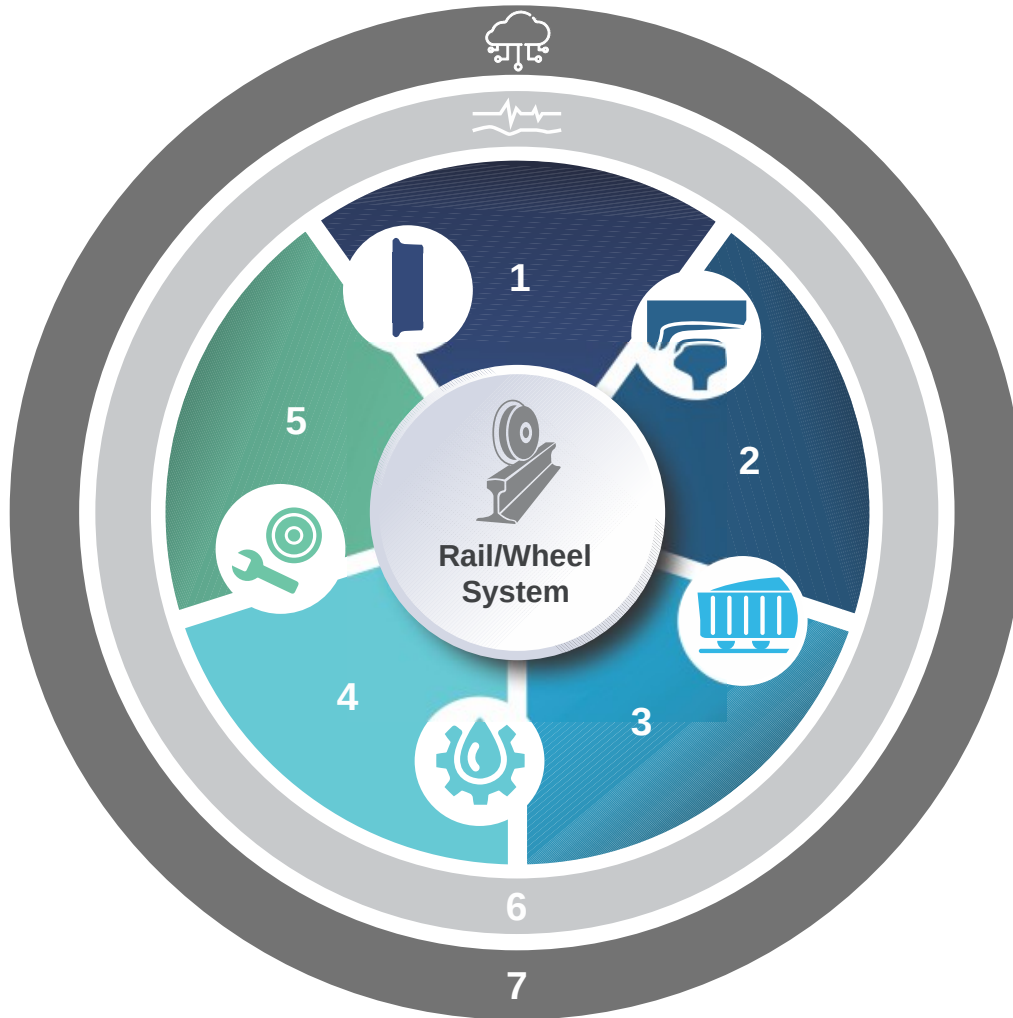
Mitigating Rail Damage

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- 1 Material and Joining
- 2 Profiles
- 3 Track Quality
- 4 Friction Management
- 5 Rail Treatment / Rail Maintenance
- 6 Measurement Technology
- 7 Data Handling Technology

Mitigating Wheel Damage

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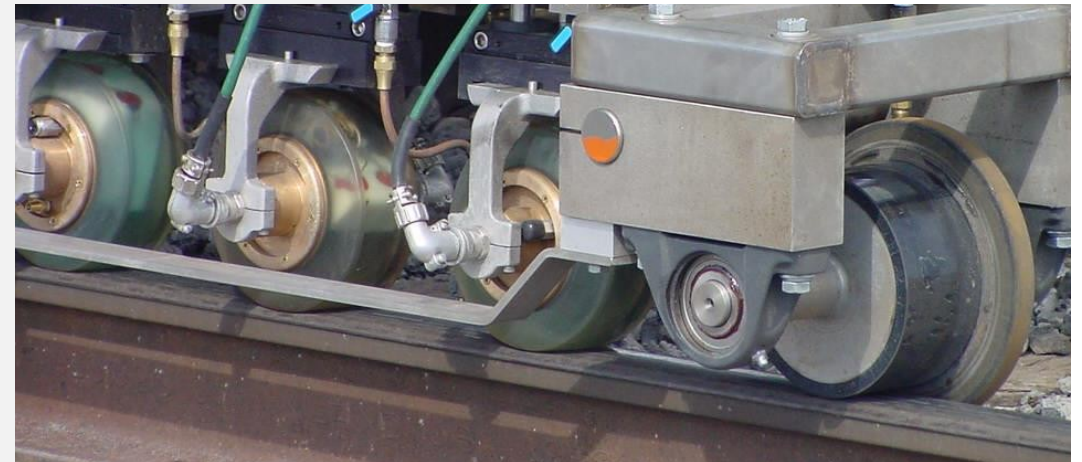
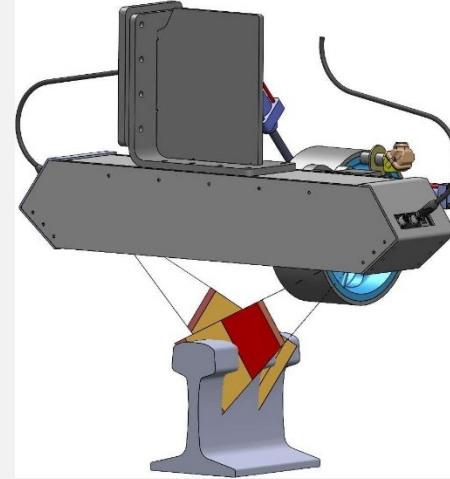


- 1 Wheel Materials
- 2 Wheel/Rail Profiles
- 3 Vehicle Condition
- 4 Friction Management
- 5 Wheel Maintenance
- 6 Measurement Technology
- 7 Data Handling Technology

Measurement Technology

- Transversal Profile
- Longitudinal Profile / Roundness
 - Periodic and singular defects
- Visual Inspection
- Vision Technology
 - Surface features
- Electromagnetic Technology
 - Near surface damage
- Ultrasound Testing
 - Internal damage
- Hand tools, vehicle integrated, wayside systems

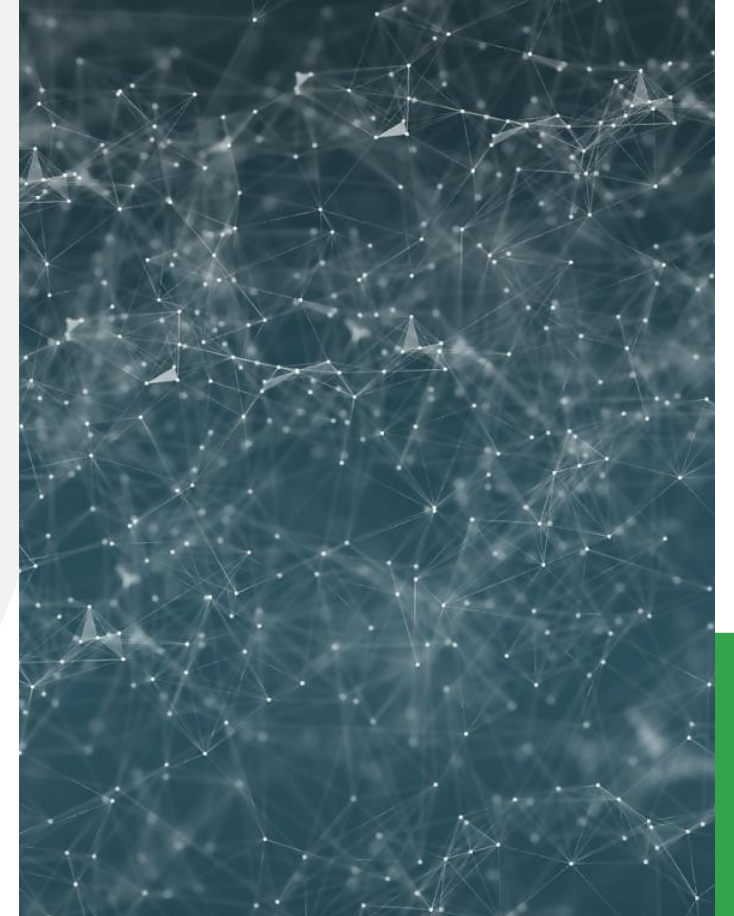
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Data Handling & Management

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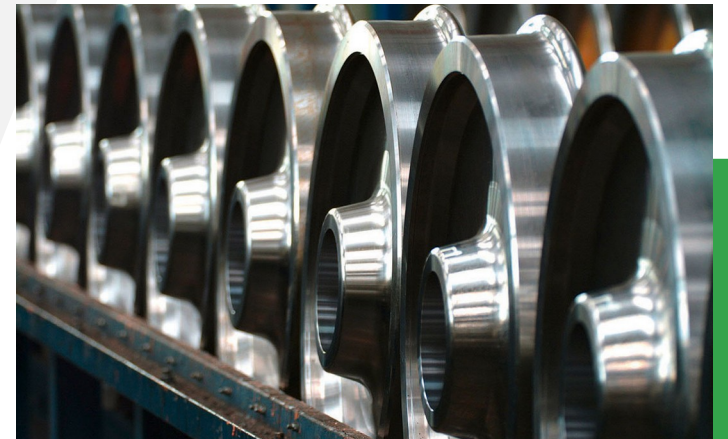
- Requirements
 - Import from various data sources
 - Data integrity, integration and analysis
 - Data synchronisation between machines and measurement devices
 - Cyber-Security (Data, machines, measurement devices, back office)
- Visualisation and Reports
- Work order creation and transfer to maintenance machine



System Damage Mitigation – I

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- Advanced Materials
 - Optimised material structure
 - Reduced wear and damage (both)
- Rail Welding
 - Prevention of premature damage on welds
- Optimised Profiles
 - Reduced contact stresses
 - Improved steering (curves) and stability (tangent)



System Damage Mitigation - II

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- Track and vehicle condition
 - Dynamic system interaction
 - Mutual beneficial or detrimental effect
- Friction Management
 - Optimised friction between wheel and rail (TOR / GF)
 - Improved steering, reduced contact stresses / damage



Source: LB Foster

System Damage Mitigation - III

- Rail Maintenance
 - Grinding and Milling
- Wheel Maintenance
 - Wheel Turning
- **The only measure that can remove damage and not delay the formation!**

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Source: Danobat

Rail Steel Grades

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AREMA Chapter 4 Rail (2021)

EN 13674-1:2011

Carbon Standard Strength

Low Alloy Standard Strength

Carbon Intermediate Strength (HH)

Low Alloy Intermediate Strength

High Strength (HH)

Low Alloy High Strength (HH)

Steel grades according to AREMA Chapter 4 Rail (2021) and EN 13674-1:2011

	Chemical composition (%)						Mechanical data		
grade	C	Si	Mn	P _{max}	S	Cr	R _m [Ksi] min	Elong. [%] min	Hardness [HB]
R260	0.62-0.80	0.15-0.58	0.70-1.20	0.025	0.08-0.025	≤0.15	127	11	260-300
SS	0.74-0.86	0.10-0.60	0.75-1.25	0.020	0.020	0.30	142.5	10	310
LA	0.72-0.82	0.10-0.50	0.80-1.10	0.020	0.020	0.25-0.40	142.5	10	310
IS	0.74-0.86	0.10-0.60	0.75-1.25	0.020	0.020	0.30	155	10	350
IH	0.72-0.82	0.10-1.00	0.70-1.25	0.020	0.020	0.40-0.70	147	8	325
R350HT	0.72-0.80	0.15-0.58	0.70-1.20	0.020	0.025	≤0.15	170	10	350-390
R350LHT	0.72-0.80	0.15-0.58	0.70-1.20	0.020	0.025	≤0.30	170	10	350-390
HH	0.74-0.86	0.10-0.60	0.75-1.25	0.020	0.020	0.30	171	10	370
LH	0.72-0.82	0.10-1.00	0.70-1.25	0.020	0.020	0.40-0.70	171	10	370
R370CrHT	0.70-0.82	0.40-1.00	0.70-1.10	0.020	0.020	0.40-0.60	185	10	370-410
R400HT	0.90-1.05	0.20-0.40	1.20-1.30	0.020	0.020	≤0.30	185	10	400-440

Several Hypereutectoid grades and experimental Bainitic grades

Wheel Steel Grades

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AAR M-107/M-208 (2020)

EN 13262:2021

Wheel Steel grades according to AAR M-107/M-208 and EN 13262:2021

	Chemical composition (%)						Mechanical data		
grade	C	Si	Mn	P _{max}	S	Cr	R _m [MPa] min	Elong. [%] min	Hardness [HB]
ER6	0.48	0.40	0.75	0.02	0.015	0.30	780-900	15	225
ER7	0.52	0.40	0.80	0.02	0.015	0.30	820-940	14	235
ER8	0.56	0.40	0.80	0.02	0.015	0.30	860-980	13	245
ERS8	0.57	1.10	1.10	0.02	0.015	0.30	900-1020	13	250
ER9	0.60	0.40	0.80	0.02	0.015	0.30	900-1050	12	255
Class L	0.47	0.15-1.00	0.60-0.90	0.03	0.005-0.04	0.25	-	-	197-277
Class A	0.47-0.57	0.15-1.00	0.60-0.90	0.03	0.005-0.04	0.25	-	-	255-321
Class B	0.57-0.67	0.15-1.00	0.60-0.90	0.03	0.005-0.04	0.25	-	-	302-341
Class C	0.67-0.77	0.15-1.00	0.60-0.90	0.03	0.005-0.04	0.25	>1100	>14%	321-363
Class D	0.67-0.77	0.15-1.00	0.60-0.90	0.03	0.005-0.04	0.25	>1100	>14%	341-415

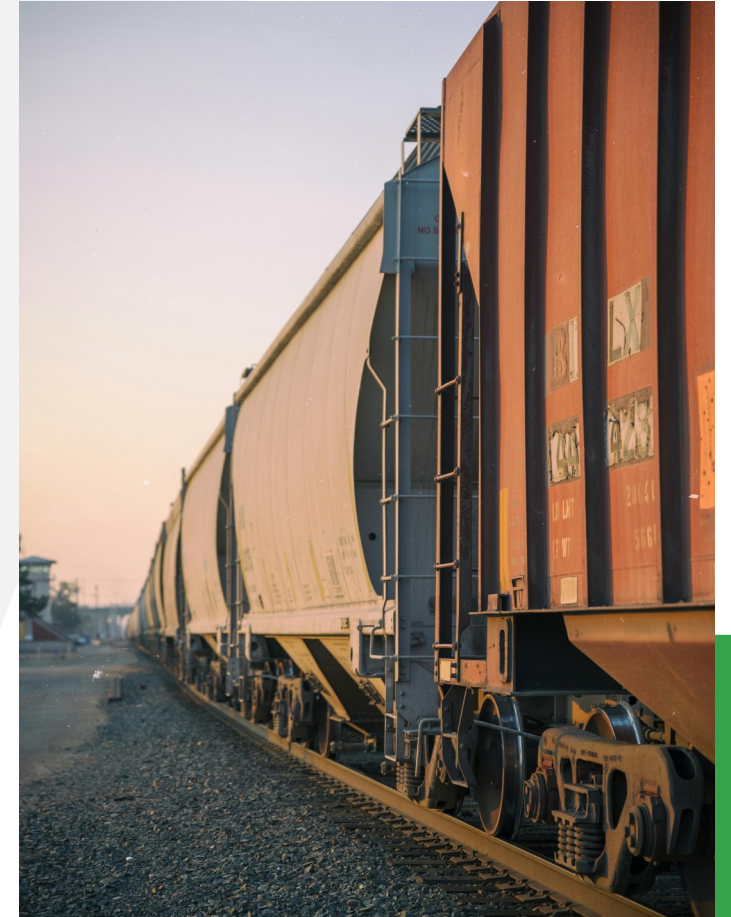
No tension testing required

Microalloying Additions: Nb, V, Mo, Ni

Summary - I

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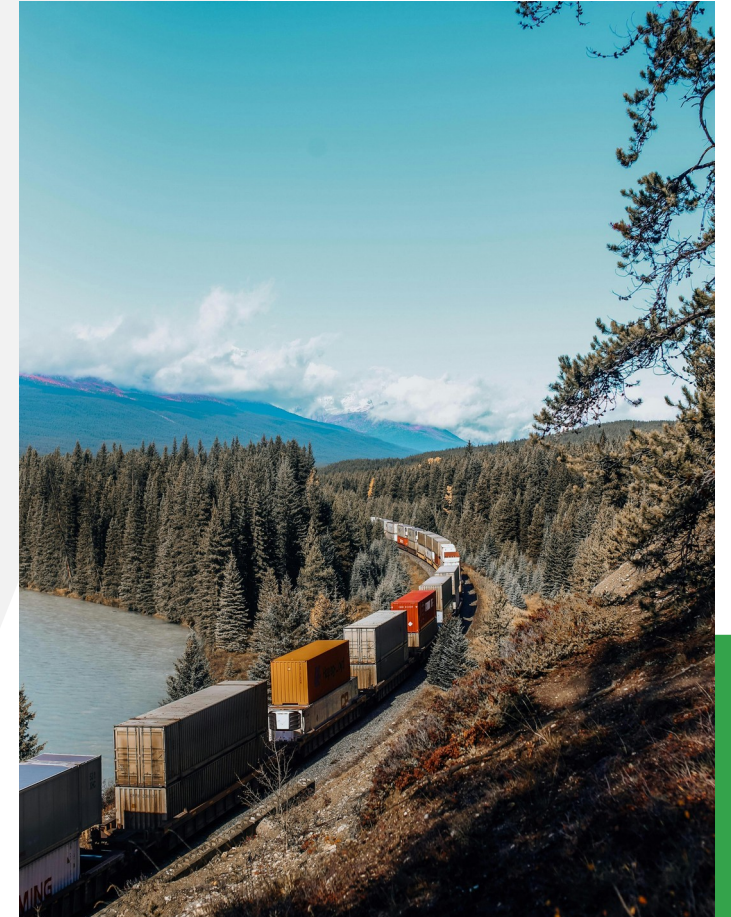
- Rail and wheel material
 - Typical rail/wheel steel: pearlitic steel
 - Lamella material structure
- Rail / wheel damage types
 - Plastic deformation, wear, cracks, thermal damage



Summary - II

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- Controlling damage
 - Measurement Technology and data handling
 - Material selection, rail welding, wheel and rail profiles, track and vehicle condition, friction management, wheel and rail maintenance
 - Ensure maximum rail (track) and wheel (vehicle) life





Thank you for your attention!

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